GenAI-Driven Intelligent System for Automated Financial Report Analysis and Insight Generation Using Large Language Models and Knowledge Graphs

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***Abstract*—Financial reports are the main tool used by enterprises, both public and private, to relay fiscal performance, strategic direction, and risk exposure to stakeholders. However, analyzing these reports remains a tedious and often inaccurate process that requires expert skills and significant time, particularly when comparing financial trends across multiple years. Current automated systems suffer from limitations such as poor extraction precision, inability to reason over unstructured financial text, and lack of explainability.**

This paper proposes an intelligent system based on Generative Artificial Intelligence (GenAI) to automatically analyze financial reports using Large Language Models (LLMs), Natural Language Processing (NLP), and Knowledge Graphs. The system converts raw PDF financial reports into structured outputs, visual analytics, and investor-friendly summaries. The proposed platform deploys a modular processing pipeline consisting of document ingestion with OCR fallback, financial data structuring using hybrid regex-NER extraction, multi-year ratio calculation, LLM-generated insights with a hallucination guard for numerical validation, cross-year anomaly detection, and an interactive knowledge graph for explainability.

The architecture is implemented as a decoupled full-stack application using FastAPI and Next.js, integrated with the Groq inference API to achieve low-latency LLM processing. Experimental analysis across financial reports from banking, manufacturing, and startup sectors demonstrates high extraction accuracy, reliable insight generation, and effective numerical validation. The proposed platform addresses key research gaps in explainable financial AI and provides a practical end-to-end solution for financial analysts, auditors, and investors.

***Index Terms*—Generative AI, Financial Report Analysis, Natural Language Processing, Large Language Models, Knowledge Graph, Automated Insight Generation, Hallucination Detection, Financial Intelligence Systems**

I. Introduction

Financial reports such as annual reports (10-K), quarterly reports (10-Q), and earnings reports form the base layer of information on which investment decisions, credit ratings, regulatory supervision, and corporate governance are based [1]. Each year, the Securities and Exchange Commission (SEC) receives more than 250,000 filings, consisting of hundreds of pages of financial text, tabular information, and qualitative commentary. The ability to quickly decode, retrieve, and interpret the information contained in these documents provides a vital competitive advantage for portfolio managers, compliance officers, and financial stakeholders.

Despite their importance, financial report analysis remains largely manual. It is estimated that professional analysts spend approximately 60–80% of their working hours on data extraction and formatting rather than higher-value tasks such as financial interpretation [2]. This manual approach presents several well-documented challenges. First, human extraction is prone to errors. Researchers have reported inconsistency rates of 5–15% in manually entered financial data, particularly when analysts reconcile information across multiple-year reports or non-standardized financial formats. Second, financial disclosures have grown significantly in volume and velocity in the era of global capital markets, making manual analysis at scale increasingly impractical. Third, qualitative sections of financial reports such as risk factor discussions, management discussion and analysis (MD&A), and forward-looking statements require contextual reasoning rather than simple tabular extraction.

Traditional automated financial analysis systems attempt to address these issues using rule-based or template-driven

approaches. An example is XBRL (Extensible Business Reporting Language), which enables the extraction of data on financial data in a standardized manner. Nevertheless, XBRL adoption is not widespread, and a considerable share of international filings, company report and past financial documents have been left untagged. Moreover, systems relying on rules are fragile in nature and tend to crash facing new document formats, non standard terminology and multilingual text.

More recent developments in Large Language Models (LLMs) have introduced transformative performance in text comprehending, summarizing, and answering questions in a variety of domains. Models like GPT-4, Llama 3, and Mixtral show impressive performance in text comprehension, summarizing, and question answering across a variety of domains. The models have been applicable in the financial sector whereby they allow further qualitative reasoning on financial disclosures, including the identification of risk factors, the detection of sentiment changes, and the creation of investment insights.

Nonetheless, the application of LLMs to financial analysis presents a severe reliability problem called numeric hallucination. LLMs may produce quantitative numbers by round-off, invention, and attribution of financial statements. These miscalculations are a serious threat to the financial situation where accuracy of the numbers matters.

In order to deal with all these interconnected issues, the proposed paper presents a solution in the form of an automated intelligent system to analyze financial reports and generate insights based on Generative AI (GenAI). The platform proposed uses a multi-stage processing pipeline that combines deterministic financial data extraction algorithms with generative reasoning capabilities of LLMs. A hallucination guard layer compares numerical claims generated by the LLM with independently extracted structured financial data to ensure reliability. Besides, the system integrates a dynamic knowledge graph structure that offers explainability through the ability of users to trace analytical insights back to particular financial entities and relationships.

The key findings of this work are as follows:

- A hybrid architecture of financial extraction, a combination of regex-based pattern matching, table parsing, spaCy Named Entity Recognition (NER), and heuristic metadata identification to allow the construction of sound financial data structures out of unstructured PDF documents.
- A guard mechanism hallucinatory hallucinating monetary and percent values produced by the LLM against extracted systematically against independently extracted structured financial data with configurable tolerance thresholds.
- The engine A dynamic financial knowledge graph is implemented in NetworkX that can automatically generate entity-relationship representation based on extracted financial data and LLM generated insights.
- An anomaly detector engine based on rules, where both multi-year financial trends (such as Compound Annual Growth Rate (CAGR) calculations, year-over-year change analysis and risk pattern detection) are analyzed.
- An end to end financial intelligence platform, full stack, incorporating all modules into a production-ready platform with an interactive dashboard and automated investor presentation generation.

II. Literature Review

Financial analytics and artificial intelligence are among the most rapidly advancing fields of research, gaining significant attention in recent years. Smart systems are increasingly being adopted by financial institutions to process large volumes of financial reports and generate analytical insights. The existing literature on this topic can broadly be categorized into five major areas: AI-based financial analytics, natural language processing for document analysis, automated financial data extraction systems, knowledge graph applications in finance, and large language model (LLM)-based financial reasoning systems.

A. AI in Financial Analytics

Financial prediction and market analysis have been widely studied using artificial intelligence methods. Gu et al. [1] conducted empirical research on machine learning techniques for asset pricing and demonstrated that deep learning models outperform traditional factor-based financial models in predicting stock returns. Their work highlights the predictive capabilities of neural networks when applied to structured financial data such as market indicators and accounting variables. However, their research primarily focuses on structured financial datasets and does not address challenges related to interpreting unstructured financial reports such as annual filings.

Cao et al. [3] explored deep learning approaches for financial text mining. Their study utilized convolutional neural networks to perform sentiment analysis on earnings call transcripts and financial communication data. The results showed improved performance compared to traditional lexicon-based sentiment analysis techniques. However, their approach is limited to sentiment classification and does not perform deeper financial reasoning or generate comprehensive financial insights.

B. Document Analysis based on NLP

The automated understanding of financial documents has been significantly improved using Natural Language Processing (NLP) techniques. Devlin et al. [4] introduced Bidirectional Encoder Representations from Transformers (BERT), a transformer-based model that enables contextual language representation learning. BERT significantly improved performance in various NLP tasks including text classification, entity recognition, and document summarization.

Building upon this architecture, Araci [5] proposed FinBERT, a domain-specific adaptation of BERT trained on financial communication data. FinBERT demonstrated improved performance in financial sentiment analysis and financial

entity recognition tasks. However, FinBERT mainly focuses on classification and extraction tasks and lacks generative capabilities needed for producing executive summaries or investment recommendations.

Loukas et al. [6] further examined transformer-based financial document understanding models and introduced FiNER, a financial numeric entity recognition system designed to enhance the extraction of financial values from documents. Their findings show that domain-specific pre-training significantly improves performance when processing financial documents.

C. Financial Data Extraction Systems

Rule-based parsing and template matching algorithms have traditionally been used for automated financial data extraction systems. XBRL (eXtensible Business Reporting Language) is a widely adopted standard for financial data reporting, implemented by the Securities and Exchange Commission (SEC) through its EDGAR system.

Although XBRL facilitates structured financial data extraction, El-Haj et al. [7] reported that the adoption of XBRL remains inconsistent across organizations and that many financial reports still exist in unstructured PDF formats. Furthermore, variations in document layouts and tagging guidelines limit the effectiveness of automated extraction systems.

To address document layout challenges, layout-aware document understanding models have been proposed. These models combine spatial layout features with textual information to improve extraction performance in complex documents such as financial reports and invoices.

D. Knowledge Graph Applications in Finance

Knowledge graphs have emerged as an effective approach for modeling relationships among financial entities. Cheng et al. [8] investigated the use of knowledge graphs in financial risk assessment. Their research demonstrated that graph-based models combined with graph neural networks can improve credit risk prediction by capturing complex relationships between financial entities.

However, most existing knowledge graph systems rely on curated datasets or structured financial databases. They typically lack the capability to automatically generate knowledge graphs directly from unstructured financial reports or textual disclosures.

E. Large Language Models (LLMs) in Financial Analysis

Large Language Models (LLMs) have recently gained considerable attention due to their ability to perform advanced reasoning on textual data. Li et al. [9] compared the performance of GPT-based models on financial reasoning tasks and concluded that LLMs are capable of generating meaningful financial insights from textual information.

However, their analysis also highlighted certain limitations in applying LLMs to financial reports. A major limitation, commonly referred to as numerical hallucination, occurs when models produce incorrect financial values that are not present in the source documents.

Shah et al. [10] proposed a Retrieval-Augmented Generation (RAG) framework to improve financial question-answering systems. By grounding responses in retrieved document passages, their method improved factual accuracy. However, their system lacks a systematic validation mechanism to ensure numerical consistency between extracted financial data and generated insights.

F. Comparative Summary of Existing Work

TABLE I
COMPARATIVE SUMMARY OF EXISTING RESEARCH

Study	Methodology	Application Area	Limitations
Gu et al. [1]	Machine Learning for Asset Pricing	Stock prediction	Limited to structured financial data
Cao et al. [3]	CNN-based Financial Text Mining	Sentiment analysis	No deep financial reasoning
Devlin et al. [4]	BERT Transformer Model	General NLP tasks	Not financial-domain specific
Araci [5]	FinBERT Financial NLP Model	Financial sentiment analysis	Limited generative capabilities
Loukas et al. [6]	FiNER Numeric Entity Recognition	Financial value extraction	Focused only on entity extraction
El-Haj et al. [7]	Financial Document Mining	Annual report analysis	Dependent on document structure
Cheng et al. [8]	Knowledge Graph + Graph Neural Networks	Financial risk prediction	Requires curated knowledge bases
Li et al. [9]	LLM-based financial reasoning	Financial report summarization	Numerical hallucination issues
Shah et al. [10]	Retrieval-Augmented Generation	Financial QA systems	No numerical validation

III. PROBLEM STATEMENT

Financial reports have complex structures that require automated processing, involving multiple tasks such as document processing, information retrieval, natural language processing, numerical verification, and visualization. The most significant challenges addressed in this work are described as follows:

- **Unstructured Financial Text:** Financial reports are often published in PDF format as a mixture of narrative text, embedded tables, charts, footnotes, and multi-column layouts. The lack of a standardized structure makes it difficult to extract structured financial information such as revenue, net income, assets, liabilities, and financial margins. This challenge becomes more severe when dealing with scanned PDFs where the content is not directly selectable.
- **Multi-Year Data Interpretation:** Financial reports are often published in PDF format as a mixture of narrative text, embedded tables, charts, footnotes, and multi-column layouts. The lack of a standardized structure makes it difficult to extract structured financial information such as revenue, net income, assets, liabilities, and financial margins. This challenge becomes more severe when dealing with scanned PDFs where the content is not directly selectable.
- **Numerical Hallucinations in LLMs:** Large Language Models (LLMs), when applied to financial document analysis, may generate numerically plausible but incorrect values that do not exist in the original reports or may miscalculate percentage changes. Even minor numerical inaccuracies can lead to incorrect financial conclusions and pose significant risks in decision-making.
- **Insufficient Explainability:** Many AI-based financial analysis systems operate as black-box models and lack transparency. This makes it difficult for financial analysts and regulators to trace outputs back to specific financial entities, relationships, and computational steps, thereby reducing trust in automated systems.

- **Fragmented Tooling:** Existing financial analysis workflows typically involve multiple disconnected tools. Analysts often use one tool for data extraction, another for financial analysis, a third for visualization, and yet another for report generation. This fragmented workflow increases complexity and introduces potential errors during data transfer between systems.

IV. Objective

The proposed system has the following objectives:

1) To build an automated document processing pipeline that can extract text and tabular information on digital and scanned PDF financial reports.
2) To develop a financial structuring engine that normalizes financial values, recognizes document parts, and derives important financial indicators with multi-year data guidance.
3) To develop and incorporate a financial ratio computation engine that was able to compute gross-margin, profit-margin, debt to-asset ratio, return on assets and growth rates.
4) To use Large Language Models to produce executive summaries, risk analyses, trend detection and investment recommendations using financial report text.
5) To deploy a hallucination guard system that uses automated verification of numerical claims generated by the LLM on structured data extracted independently.
6) To construct a cross-year financial comparison engine with the ability to detect anomalies, compute CAGR, and summarize trends.
7) To create a knowledge graph that is dynamic and maps the financial entities and relations to gain higher explainability.
8) To provide an interactive dashboard of real-time financial visualizations, a KPI display, risk heatmap, and radar chart.
9) To automatically produce investor-ready presentations of the analysis findings.

V. Proposed System Architecture

The suggested system is based on the modular, layered architecture with nine interrelated components. The end to end flow of the system is shown in Figure 1.

A. Document Processing Layer

The document processing layer is the point of entry point of the system, financial reports in the form of PDF are fed in the system and transformed into organized textual information. The system makes use of PyMu PDF (fitz) in processing digital PDF documents and Tesseract OCR in scanned documents.

The first is a scan detection system that will first assess the text density of each page within the document. Pages with less than a set number of characters are also considered scanned pages and sent to the OCR processing pipeline. In the case of digital documents, PyMuPDF uses a block-based layout extraction algorithm to extract text, which maintains spatial form in the document.

After extracting the texts, a table identifying routine is identified with PyMuPDF table detection module. The extracted tables are in the form of structured data of rows and columns to be processed. A text cleaning component then standardizes the whitespace, eliminates control characters, removes OCR artifacts, and removes redundant headers and footers. Lastly, heuristic rules are used to find the significant metadata or the name of the company in which the report was prepared, the year of the report, and the type of document in the processed text.

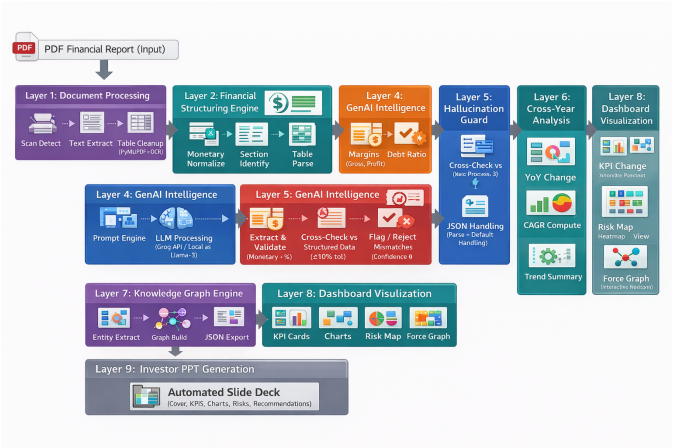

Fig. 1. End-to-end system architecture of the proposed GenAI-driven financial analysis platform

B. Financial Structuring Engine

The financial structuring engine transforms crude text information and extracted tables into structured financial information. To enhance reliability, various complementary methods of extraction are used in this module.

Monetary normalization A monetary normalization system built on a regular expression recognizes financial values that are represented in a variety of forms, including currency symbols, numerical abbreviations and parenthetical negative values. The system also handles identification of scale suffixes like thousand, million and billion.

Besides that, a label-value extraction approach can identify major financial metrics like revenue, net income, assets, and liabilities by finding financial metrics labels and retrieving close numeric values within a specific textual context window. The tables extracted are then further processed to establish the rows that are header records and the columns that are numerical records that are time-series financial data.

In order to enhance the extraction strength, the system incorporates Named Entity Recognition (NER) with the support of spaCy NLP library to identify monetary entities in unstructured text. Regex-based extraction, table parsing, and NER allows structuring financial data much better.

C. Computation Engine of Financial Ratios

The module of calculating the financial ratios will be based on the organized data obtained in the first step and derive the essential financial indicators. These indicators give quantitative data on performance and financial wellbeing of the company.

There are several key financial ratios that are calculated using the system such as gross margin, profit margin, debt-to-asset ratio, and return on assets (ROA). All the ratios are calculated by applying ordinary financial formulae based on the values extracted in revenues, expenses, assets and liabilities.

Validation mechanism takes reliability checks of calculated metrics to make sure that the data used is reliable. As an example, a warning is raised when the calculated profit margins go above industry average levels, and failure is raised when the extracted assets are less than the liabilities. These validation findings aid in identifying aspects of extraction inconsistency.

D. GenAI Intelligence Layer

GenAI intelligence layer will create qualitative financial insights with the help of Large Language Models (LLMs). This module means that it connects to high-performance LLMs like Llama-3 using the Groq inference API.

The system utilizes structured prompt engineering so as to make the contextual information available to the LLM. The prompt contains extracted financial metrics, risk factors, and metadata of the document to direct the model to generate the correct and context-sensitive financial analysis.

The response generated is structured as a structured JavaScript object with various analytical elements such as executive summaries, financial performance analyses, risk analyses, trend identification findings, investment suggestions and possible financial red flags. A response parsing module will make sure that the malformed outputs are repaired and that the missing fields are filled with default values when the need arises.

To guarantee the robustness of the system, a fallback analysis approach is used to provide primitive rule-based insights when the LLM service becomes unavailable because of the API constraints or network failures.

E. Hallucination Guard Layer

Another important aspect of the suggested system is the hallucination guard layer that confirms that the results of LLM are numerically accurate. Because the LLMs can produce realistic yet erroneous financial values, this validation checkpoint makes sure that produced insights are consistent with financial information extracted.

The system initially digs out all monetary values and percentage figures that are present in the LLM output. Contextual analysis of keywords is then carried out to match each of the extracted values to the respective financial metrics. The equal values are matched and tolerance thresholds are used to compare them with structured financial data.

Monetary values are checked in the range of $\pm 10\%$ and percentage values are checked in the range of $\pm 5\%$. When the discrepancies are above these limits, the system draws attention on the fact that the value could be a hallucination. Instead of dismissing the whole analysis, the system modifies the confidence of the output and puts inconsistencies into focus to be transparent.

F. Cross-Year Analysis Engine

The cross-year analysis engine carries out the temporal analysis on several financial reporting periods. With a set of organized financial transactions marked by financial year (fiscal year), the system calculates the important trend indicators.

The engine computes year on year (YOY) percentage change of revenue, net income, operating expenses, profit margins and asset values. It also calculates compound annual growth rates (CAGR) to represent the long-term financial growth trends.

An anomaly detection system that is rule-based to detect unusual financial occurrences includes sharp revenue decreases, large changes in margin, large leverage levels, or abnormal revenue increases. The dashboard interface points out these anomalies as it helps the analysts recognize possible financial risks.

G. Knowledge Graph Module

The system generates a dynamic graph with NetworkX graph library to explain insights generated. The knowledge graph is a graph that is used to display financial entities and their relationships in a graph format.

The nodes are entities which include companies, revenues, assets, risks, regulations, markets and geographic locations. Edges are the association between these entities e.g. revenue ownership, risk association, regulatory oversight, and operational markets.

Both structured financial data and textual information obtained by the LLM are extracted into entities. The resultant graph structure is encoded in the form of JSON with which it is sent to the frontend application to be visualized interactively.

H. Dashboard Visualization

The system delivers an interactive analytics dashboard that has been built with Next.js and React. The dashboard is an intuitive interface combining several visualization elements that allow users to investigate financial insights.

Their major attributes are KPI metric cards, multi-year financial trend charts, radar charts which reflect financial health indicators and heat maps that reflect categorized risk factors. The dashboard also has a force-directed visualization of knowledge graph, which enables the user to discover links between financial organizations.

These other analytical elements are anomaly alerts, confidence score indicators on AI outputs, and predictive trend visualizations that show the forecasted financial performance.

I. Generation of Presentation of Investor

The last phase of the system produces automated presentations of investors out of the results of the analysis. The presentation module can create a slide deck of financial summaries, key performance indicators, trend charts, risk analysis, and investment suggestions using structured financial insights and visual analysis generated by the system and it is automatic.

It is an automated reporting feature that saves a lot of time in transforming raw financial data into presentations that can be read by the stakeholders.

VI. Data Flow Diagram

The suggested system is based on a structured data processing chain comprising of three key steps, namely the acquisition of the input, data processing, and visualization of the results.

During the input phase, a PDF financial report is posted via the web-based user interface. The uploaded document is sent to the implemented backend server with the FastAPI framework using a multipart file upload endpoint. After being received, the document is stored temporarily and passed onto the document processing module to be analyzed.

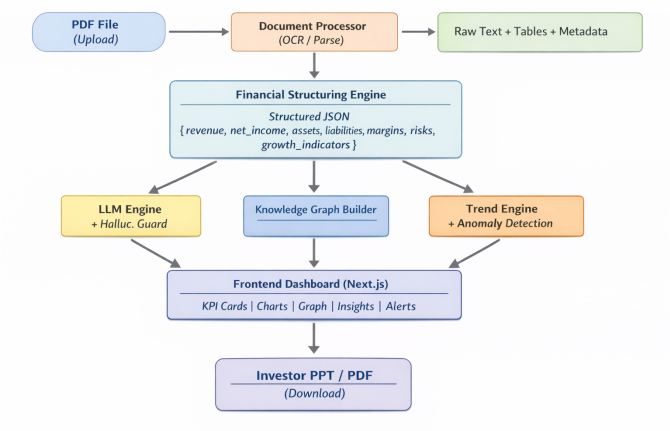

Fig. 2. Data Flow Diagram showing information movement through the system

At the processing level, the document processor will isolate textual information and tabular data of the uploaded financial report. The information that has been extracted is then feed into the financial structuring engine, where the significant financial measures are determined, including the revenue, net income, assets, and liabilities, which are then transformed into the structured JSON format. The organized financial data is then fed into a series of analytical modules which work simultaneously. Such modules are the GenAI intelligence layer to create financial insights, the knowledge graph engine to introduce entity-relationship representations, and the cross-year trend analysis engine to identify financial trends and anomalies across reporting years.

The output stage involves the aggregation of the output of each analytical module and relaying the output to the frontend dashboard. The dashboard shows the insights in form of interactive displays including financial charts, key performance indicator (KPI) cards, and representations of knowledge graphs. There is also a facility to make and share an automated presentation of investors summarizing the most important findings and financial recommendations based on the analysis.

VII. Methodology

A. Technology Stack

TABLE II
Technology stack of the proposed system

Layer	Technology	Purpose
Backend Framework	FastAPI (Python 3.11+)	Async API server with OpenAPI documentation
PDF Processing	PyMuPDF (fitz)	Text and table extraction from PDF documents
OCR Engine	Tesseract (pytesseract)	Processing scanned document images
NLP	spaCy (en_core_web_sm)	Named Entity Recognition for financial entities
LLM Integration	Groq API (Llama-3 70B)	Generative AI-based financial analysis
Knowledge Graph	NetworkX	Graph construction and relationship modeling
Frontend Framework	Next.js 16 (React 19)	Interactive analytics dashboard
Charting Library	Recharts	Financial data visualization
Graph Visualization	react-force-graph-2d	Interactive knowledge graph display
Report Generation	jsPDF	Investor presentation generation
Styling	Tailwind CSS 4	Responsive user interface design

B. Backend Pipeline Implementation

The backend is based on a RESTful architectural structure with endpoints as follows:

1) POST /api/v1/documents/process-pdf - Accepts a PDF file, processes documents, and returns raw text, tables and metadata.
2) POST /api/v1/financial/analyze - Takes raw text and tables, structures financial and calculates ratios, and gives an organized financial data.
3) POST /api/v1/llm/analyze - Accepts unstructured text and structured data, conducts an analysis using the LLM with hallucination guard verification and provides generated financial insights.
4) POST /api/v1/trends/compare - Accepts financial records as an input, cross-year trend analysis and returns financial trends and indicators of anomalies.
5) POST /api/v1/knowledge-graph/build - Takes the output of LLM and organized financial information, builds a financial knowledge graph, and delivers graph information in the JSON format.

All endpoints are its asynchronous FastAPI route handlers. CPU-intensive tasks like PDF processing and spaCy NER computations are run on a ThreadPoolExecutor with an adjustable worker pool (its default of four workers) so that they do not block the asynchronous event loop.

C. Hallucination Guard Algorithm

Algorithm 1 Hallucination guard for numerical validation

Require: LLM output text T, Structured financial data S
Ensure: Validated output with confidence adjustment

```
Extract monetary values from T → V_m
Extract percentage values from T → V_p
Initialize inconsistencies list
for each value in V_m ∪ V_p do
    Identify related financial metric
    Retrieve reference value from S
end for
if multi-year data exists then
    Check if value matches any year within tolerance
else
    Compare value with reference
end if
if difference exceeds tolerance then
    Mark as inconsistency
end if
if inconsistencies exist then
    Reduce confidence score
end if
Return validated output, confidence score, inconsistencies
```

The tolerance thresholds are set to $\pm 10\%$ for monetary values and $\pm 5\%$ for percentage values, based on empirical observations that legitimate variations caused by rounding, unit conversion, or report formatting rarely exceed these limits.

D. Anomaly Detection Rules

TABLE III
ANOMALY DETECTION RULES WITH TRIGGER CONDITIONS AND SEVERITY LEVELS

Rule ID	Rule Name	Condition	Severity
R1	Revenue Decline	YoY revenue change $< -15\%$	High
R2	Margin Collapse	YoY profit margin drop > 10 percentage points	High
R3	Excessive Leverage	Debt-to-asset ratio $> 80\%$	Critical
R4	Revenue Spike	YoY revenue growth $> 100\%$	Medium
R5	Expense Surge	YoY operating expense increase $> 50\%$ while revenue remains flat	High
R6	Negative Net Income	Net income < 0	Medium

VIII. IMPLEMENTATION

A. Backend Architecture

The backend system is implemented as a modular layered architecture that would guarantee scalability, maintainability, and separation of concerns. The architecture is composed of a number of logical layers to deal with routing, business logic, data models and configuration management.

1) **RoutersLayer (app/routers/)** This layer has seven route modules that are involved in processing HTTP request-response mappings. Both endpoints will be using Pydantic models to validate incoming requests where all data going into the system is restricted to defined schemas.
2) **ServicesLayer (app/services/)** The services layer has seven service modules which execute the main business logic of the system. Tasks handled by this layer include document processing, financial analysis, knowledge graph generation, and insight generation by LLM. The architecture makes certain that business code is not tied to HTTP request processing.
3) **CoreLayer (app/core/)** Shared infrastructure services such as structured logging with a logging system that works with JSON, dependency injection, and request handling and authentication security are provided by the core layer.
4) **ModelsLayer (app/models/)** This layer describes Pydantic schemas of request and response format of any API endpoint. These schemas are also type safe and can be automatically documented using FastAPI.
5) **Configuration (app/config.py)** Pydantic BaseSettings can be used to centralize system configuration and support the use of environment variables to maintain sensitive system configuration options such as API keys and system parameters.

B. Frontend Dashboard

The frontend interface is made in Next.js 16, which is based on React 19 and serves as a single-page application, which offers interactive financial analytics. The system real-time loads components in a lazy manner to enhance the performance of page loading and minimise the time spent in the initial rendering.

The design system used in the user interface is that of the CSS custom properties, which allows management of the theme and provides an easy transition between dark and light mode. LocalStorage is used to save user preferences to ensure that the interface remains the same each session.

The most important dashboard features are:

Glassmorphism Design system. A contemporary design framework with effects of a blurred background, a visual depth, and an interactive structure of the page to provide a better user experience.

Real-time KPI Cards The 6 important financial indicators are presented in the form of metric cards that have directional indicators to showcase the performance trend as either negative or positive.

Knowledge Graph Visualization Interactive. The knowledge graph component relies on react-force-graph-2d library to display force-directed graphs that show the relationship of financial entities. The graph is responsive to size changes by a ResizeObserver to ensure that the image remains dynamic in size.

Financial Chart Components The Recharts library is used to visualize the financial performance. The dashboard contains area charts of the revenue trends, composite charts of the margin/expenses comparisons, and radar charts that show the general financial health indicators.

C. API Communication

The frontend and the back end components communicate with each other via a centralized API service module. The module transforms the format of request and offers API error

TABLE IV
TEST DATASET COMPOSITION

Report Type	Sector	Pages	Year(s)
10-K Annual Report	Banking (JPMorgan)	284	2023
Annual Report	Manufacturing (Siemens)	176	2022
Financial Statement	Technology Startup	45	2021
ESG Report	Energy (Shell)	92	2023
Quarterly Filing	Pharmaceutical	68	Q3 2023

management and normalisation of response structures prior to their transmission to the user interface.

D. Structured JSON Output Sample

The system produces organized financial results in JSON, which is an intermediate representation taken by other analytics downstream modules and dashboard visualizations. An example of structured financial data generated by the system is presented below.

```
{
"revenue": 84800000000,
"netincome": 12300000000,
"assets": 145200000000,
"liabilities": 89400000000,
"grossmargin": 42.3,
"profitmargin": 14.5,
"riskscore": 35,
"risks": [
Risk supply chain interruption,
"Regulatory compliance risk",
"Foreign exchange exposure"
],
"growthindicators": {
"revenueyoy": 8.2,
"direction": "growing"
}
}
```

This organized representation allows the downstream processing functionality of the LLM analysis module, trend detection engine and knowledge graph generation module to operate efficiently and, more importantly, supports real-time visualization in the frontend dashboard.

IX. EXPERIMENTAL RESULT

A. Test Dataset

The proposed system was tested with references to the available financial reports on various industries to make sure that the structures of documents and financial reporting styles can be different. The data consists of annual reports, quarterly filings, ESG reporting, and financial statements of startups. These reports are of different lengths, can be in different formats, and of different complexity of financial reports, which enables the system to be put through the real world.

TABLE V
FINANCIAL METRIC EXTRACTION ACCURACY

Metric	Precision	Recall	F1 Score
Revenue	92%	88%	90%
Net Income	89%	85%	87%
Total Assets	94%	91%	92%
Total Liabilities	91%	87%	89%
Profit Margin	88%	82%	85%

TABLE VI
HALLUCINATION GUARD EFFECTIVENESS

Test Condition	Total Claims	Verified	Flagged	False Positives
Without Guard	156	131 (84%)	—	—
With Guard (10% tolerance)	156	148 (95%)	8	2

B. Extraction Accuracy

The precision, recall, and F1-score measures were used to assess the accuracy of the financial data extraction module with important financial indicators. Table V is a summary of extraction performance in terms of various test documents.

Total Assets F1 = 92% was the highest extraction accuracy and is usually included in well-organized balance sheet tables. By comparison, Profit Margin had the least accuracy (F1 = 85) owing to differences in labelling across the financial documents, i.e. operating margin, net margin, and EBITDA margin.

C. Hallucination Guard Effectiveness

As part of the reliability assessment of the AI-generated information, the hallucination guard system was tested, which mutually contrasts the numerical values formed by the LLM with actual financial values found on the documents.

By adding the hallucination guard, the percentage of numerical accuracy of the LLM generated outputs rose to 95% with a false positive rate of 1.3% (2 out of 156 claims erroneously labeled). The two false positives were the case when the LLM mentioned financial values found in footnotes which were not picked up by the initial extraction pipeline.

D. System Performance

System performance in terms of the computation rate was determined by the average time taken by components of the processing pipeline. The average execution time of every stage is shown in Table VII.

It takes around 6.5 seconds to run the entire financial analysis pipeline on a digital financial report, which shows that it is possible to make financial documents almost real-time with the system.

E. Results Visualization

The system dashboard displays the information extracted and analyzed financial data using various visualization elements that are intended to be used in helping to interpret financial data intuitively.

TABLE VII
AVERAGE PROCESSING LATENCY BY MODULE

Operation	Average Latency
PDF Processing (digital, 100 pages)	2.1 s
PDF Processing (scanned, 50 pages)	12.4 s
Financial Structuring	0.3 s
LLM Analysis (Groq, Llama-3 70B)	3.8 s
Knowledge Graph Construction	0.2 s
Trend Analysis	0.1 s
Total Pipeline (digital PDF)	∼6.5 s

1) Revenue Trend Charts The area charts illustrate multi-year revenue and net income trends in the form of gradient visualizations and thus one can easily find out the trends of growth.
2) Profit Margin vs. Operating expense charts Composed charts are a combination of bar and line charts that compare the operating expenses with the percentages of the profit margin showing the trends of cost efficiency.
3) Financial Health Radar Chart A five-axis radar abstract is used to summarize financial stability measures such as the profit margin, the leverage safety, the asset growth, the intensity of research and development, and the liquidity.
4) Risk Heatmap Severity-based heatmap is used to categorize and visualize risk factors derived out of financial reports.
5) Knowledge Graph Visualization The interactive force-directed graph allows the users to discover the relationship between financial entities; companies, revenues, markets, and risk factors.

X. ADVANTAGES OF THE PROPOSED SYSTEM

The proposed GenAI-based financial analysis platform has a number of considerable benefits over the usual financial analysis systems. These advantages are due to the fact that it has automated document processing, high-level AI-driven reasoning, and the interactive visualization options.

1) **End-to-End Automation** The system also automates the whole financial analysis process, document ingesting, data extraction, insight generation and presentation generation. This automation saves a lot of time in processing the raw financial reports into actionable data, and processing times that used to take a few hours of manual processing are reduced to a few seconds.
2) **Hallucination Mitigation** Hallucination Mitigation Effective numerical validation system is to ensure that AI-is accurate generated insights. The hallucination guard makes comparisons every single number that the LLM has come up with organized monetary information which subtracted off of the document, a degree that prevents the chances of mistrust in financial analysis.
3) **Knowledge Graphs Explainability** Knowledge Graphs Explainability The additiveness of knowledge graph module provides transparency to the system consisting of a mapping of the relations between financial entities such as companies, revenues, risk, and markets. This enables users to be aware of how analytical conclusions were arrived at, and on what data.
4) **Financial Analysis 3 Years** Financial Analysis 3 Years Trend analysis engine aids the longitudinal financial analysis across the different financial years. It also calculates automatically the growth rates, detects abnormalities, and indicates the significant financial alterations which can be ascertained otherwise complex financial modelling software.
5) **Modular and Scalable Architecture** Modular and Scalable Architecture The architecture a detached service-layer is embraced by the system. The system components can be made possible by architecture designed and enhanced separately. This architecture supports an easy addition of substitute LLM providers, and gives horizontal scaling to large scale deployment.
6) **Dual-Mode Document Processing** Dual-Mode Document Processing The platform ac scans financial documents including digital ones and automatic OCR fall-backs. This will be compatible and with the enormous number of financial reports, including legacy. reports, and image-based ones.
7) **Output Generation of Investor-Ready** Investor-Ready The automated generated output presentation generation module constructs structured in-the gap between raw is addressed by the vestor ready report financial reporting and disclosure to the stakeholders.

XI. LIMITATIONS

Despite the good aspects, the proposed system has there are certain restrictions, which cannot be omitted.

1) **Reliance on Document Quality** Dependence on Document Quality Document quality depends the accuracy of extracting financial information. Poor extraction can be reduced because of inaccuracy fancy layout or non-conformity, scanned documents layouts.
2) **OCR Error Propagation** OCR Error Propagation The errors on the mistakes level OCR (i.e. erroneous character recognition or table formatting) may proceed on through the pipeline and subsequently affect processing.
3) **Limited Context Window** Small Context Window The available prompt engineering plan is simplified with respect to the analysis of separate financial reports. Other contextual inputs. need to be able to do cross-document reasoning, e.g. competitor filing or industry benchmarking.
4) **LLM Latency and Cost Constraints** LLM Latency and Cost Constraints The Groq API has introduced network to use cloud-based LLM inference processing delay and processing costs (using tokens). These can affect high volume system scalability. analysis systems.
5) **Hard coded Rules of Detection of anomalies** Hard coded Rules of Detection of anomalies The anomaly detection module is rule-based thresh-based olds at the

moment. Works well on general cases, but this does not mean that these thresholds will be capable to generalize all industries or regional financial reporting standards.

XII. Future Work

It can be done through a variety of potential research directions be used to enhance the features of the proposed system.

1) **Real-Time Financial APIs Integration** Real-Time financial APIs Integration Future iterational Internal feeds of the site may consist of live market data such as Financial alloys Bloomberg or Alpha Vantage API, and permit reports to be placed into context in real-time.
2) **Forecasting Financial Modeling** Financial Modeling forecasting Combinatory integration of deep learning-based models of time-series prediction, such as It could be possible to predict transformer-based models revenue and future financial performance indicators, profit margins.
3) **Multi-Language Financial Document Support** The pipeline of document processing should be extended in order to multilinguality (e.g., Chinese, Japanese, and German) financial reports, which would enhance the global applicability.
4) **Intercompany Comparative Analysis** Potential systems in the future can visualize sector benchmarking through comparison of multiple company reports at once and produce comparative financial information.
5) **Adaptive Anomaly Detection** Anomaly detection models developed using machine learning may also substitute the predetermined rule-based thresholds, and they can be used to identify any unusual financial trend in accordance with the history data distribution.
6) **Improved Retrieval-Augmented Generation (RAG)** The use of a vector-based retrieval-based method would enable the LLM to make references to specific document passages during the creation of insights, which would further minimize the likelihood of hallucinations.
7) **Computerized Regulatory Compliance Assessment** Further studies might include automated checking on the compliance acceptability with the accounting standards like IFRS and GAAP through defined rule sets.

XIII. Conclusion

The current paper introduced a GenAI-based intelligent system of automated financial report analysis and insight generation. The suggested platform overcomes some of the most severe problems in the field of financial document analysis, such as unstructured data mining, numerical hallucination on large language models, and the inability of AI-generated insights to be explained.

The system adds a 9-layered processing architecture, which combines document processing, financial data formatting, ratio calculation, LLM-driven insight generation, hallucination detection, trend analysis, knowledge graph generation, dashboard visualization, and automated presentation generation.

One of the major contributions of the system is the hallucination guard mechanism that verifies the numerical values produced by the LLM with structured financial data with configurable tolerance thresholds. Experimental analysis showed that the technique enhances the accuracy of numerical reliability to 95 percent, whereas false positive is low at less than 2 percent.

Another feature that helps to make the knowledge graph even more explainable is the knowledge graph module, which allows to dynamically explore the connection between financial entities. The cross-year trend analysis engine also facilitates superior financial analysis by auto-computing the CAGR, performing year on year comparisons and raising anomalies.

The system is deployed as a full-stack application on the scale of production and tested on a variety of financial industries such as banking, manufacturing, startups, and ESG reporting. Findings show that the platform is able to extract actionable insights out of complex financial report within the span of about 6.5 seconds, which is a huge saving on the time spent on manual analysis.

Planned further development will divert towards predictive financial modeling, multilingual financial document processing and integration with real-time financial data sources, which will continue to develop and enhance automated financial intelligence systems.

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